

Discrete Logging

Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 8

AC: 3

Score: 99.81

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Codes

AI Code Tips

Description

Problem B: Discrete Logging

Given a prime P , $2 \leq P < 2^{31}$, an integer B , $2 \leq B < P$, and an integer N , $2 \leq N < P$, compute the discrete logarithm of N , base B , modulo P . That is, find an integer L such that

$$B^L \equiv N \pmod{P}$$

Read several lines of input, each containing P, B, N separated by a space, and for each line print the logarithm on a separate line. If there are several, print the smallest; if there is none, print "no solution".

The solution to this problem requires a well known result in number theory that is probably expected of you for Putnam but not ACM competitions. It is Fermat's theorem that states

$$B^{(P-1)} \equiv 1 \pmod{P}$$

for any prime P and some other (fairly rare) numbers known as base- B pseudoprimes. A rarer subset of the base- B pseudoprimes, known as Carmichael numbers, are pseudoprimes for every base between 2 and $P-1$. A corollary to Fermat's theorem is that for any m

$$B^{(-m)} \equiv B^{(P-1-m)} \pmod{P}.$$

Samples

input

Copy

```
5 2 1
5 2 2
5 2 3
5 2 4
5 3 1
5 3 2
5 3 3
5 3 4
5 4 1
5 4 2
5 4 3
5 4 4
12345701 2 111111
```

```
1111111121 65537 1111111111
```

output

Copy

```
0
1
3
2
0
3
1
2
0
no solution
no solution
1
9584351
462803587
```

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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Server Time: 2025-8-26 15:05:19